

Jinsoo Yang

Platform Engineer

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Platform and security engineer who runs LND on Kubernetes as a stateful workload. I built a reproducible LND operations lab with Helm, K3s and Prometheus on regtest and testnet, put Aperture in front of a paid API as an L402 gate, and reported bugs to Ind and aperture. Before that, 5+ years in security engineering and consulting at IBM, Deloitte and the Republic of Korea Army. CKA, CKS and CKAD certified.

LIGHTNING & PLATFORM PROJECTS

Ind-ops: LND on Kubernetes

[GitHub](#) · [write-up](#)

- Wrote one Helm chart that deploys LND as single-replica StatefulSets on K3s, run on two hosts (Lima on macOS arm64, WSL 2 on amd64). Regtest runs Bitcoin Core with two LND nodes; testnet runs one node on Neutrino.
- Kept restart persistence and disaster recovery apart: a PVC holds live wallet and channel state, the Static Channel Backup is encrypted and stored off that volume, and the wallet is unlocked by the operator after each restart.
- Started networking from default-deny NetworkPolicy with explicit allowances. P2P exposure through a NodePort is opt-in, and gRPC and REST stay cluster-internal. Pods run without a ServiceAccount token, with all capabilities dropped and RuntimeDefault seccomp.
- Wrote acceptance checks that replace the LND pod and run `helm upgrade`, then compare node pubkey, channel points, PVC identity and SCB hashes before and after.
- Routed a testnet payment through the node. Found a channel that was active but still advertised as disabled, enabled it with `updatechanstatus`, and matched the 1,005 msat fee in `fdinghistory`.

L402 payment gate for an OpenCTI-based scan service (Aperture)

[write-up](#)

- Ran Aperture as the L402 reverse proxy in front of a paid scan API, with order-based pricing served to Aperture over gRPC so the client never supplies the amount.
- Built a receipt service that looks up the invoice on the merchant LND node and accepts a payment only when it is SETTLED for the exact order amount.
- Wrote verified receipts to an outbox before changing order state, so a restart cannot lose a settled payment, and made retries on a paid order return the existing state. Ran the flow end to end on testnet.

Observability and incident rehearsal for the payment path

[write-up](#)

- Collected LND metrics with Indmon and Prometheus, with one PrometheusRule file per service and rule unit tests in `promtool`.
- Added a once-a-minute blackbox probe, a 99.5% SLO and multiwindow burn-rate alerts. In a rehearsed outage, detection went from 16 min 22 s to 5 min 14 s.
- Built a Grafana dashboard that uses the same queries as the alerts, enforced by a test.
- Built a read-only diagnosis agent on kagent: four MCP tools written in Python, no Kubernetes token for the agent, and a Role limited to `get` and `list`. An eval harness (5 scenarios, 5 runs each) raised its pass rate from 47% to 72-80%.

Hoodi Ethereum validator platform on AWS EKS

[GitHub](#)

- Provisioned the EKS cluster, VPC and IAM with Terraform and deployed the workloads through Argo CD from a separate GitOps repository.
- Built a private-first validator platform with Nethermind, Prysm, Web3Signer, Vault and PostgreSQL, keeping execution, consensus, signing, secrets and audit in separate boundaries.
- Layered controls across IAM, VPC networking, Kubernetes namespaces, private ECR delivery and Vault secret management, with CI/CD on short-lived AWS credentials.

Web3 cryptographic key management policy in OSCAL

[GitHub](#)

- Wrote an 18-section key management policy on a NIST SP 800-57, SP 800-130 and SP 800-131A baseline, with added controls for validator signing keys.
- Converted it into an OSCAL 1.2.3 catalog with stable control IDs and schema validation, then mapped the requirements against the running Vault-based validator design.

OPEN SOURCE

Ind (lightningnetwork/Ind)

- Reported a wire-decoding bug in NodeAnnouncement2 address parsing (#11211). A maintainer confirmed it and credited me as co-author on the fix (PR #11219, in review).

Aperture (lightninglabs/aperture)

- Opened PR #285, fixing a 1-in-256 flake in TestTamperedL402 (in review).
- Proposed per-outcome Prometheus counters for L402 mint and verify (#286) and implemented them in Go with tests on a fork. Proposed dedicated security event logging (#291).

Prowler (open cloud security platform)

- Added 28 security checks across Azure and GCP, including AKS, Cosmos DB, Databricks, Entra ID, Cloud SQL and Secret Manager.

OSCAL Compass (Compliance as Code project backed by NIST and CNCF)

- Contributed a GitHub Actions integration for Compliance-to-Policy workflows and fixed a Trestle KeyError bug.

SEAL Frameworks (Security Alliance, open-source blockchain security framework)

- Contributed guidance on Policy as Code, private registries and package mirrors.

EXPERIENCE

Cybersecurity Consultant, IBM

Nov 2024 - Sep 2026 | Seoul

- Designed and deployed iDMZ network architecture and OT security tooling at nuclear, semiconductor and automotive sites.
- Worked with IBM's global OT SOC to design detection logic and put monitoring use cases into operation.
- Assessed AWS environments against ISMS-P: control mapping, evidence verification and remediation planning.

Sr. Security Consultant, Deloitte Consulting Korea

Mar 2024 - Nov 2024 | Seoul

- Assessed a digital signature and certificate service, covering cryptographic controls, access management and key lifecycle.
- Reviewed AWS architecture and security controls against compliance frameworks and set remediation priorities.

Vulnerability Analysis Track, KITRI Best of the Best (BoB) 12th

Jun 2023 - Feb 2024 | Seoul

- Led development of a multi-cloud CIEM platform, designing an IAM policy normalization engine across AWS, Azure and GCP.

Signal Officer (Captain), Corps CERT, Republic of Korea Army

Dec 2021 - Mar 2023

- Managed firewalls, IDS and endpoint controls, from policy configuration through event analysis.
- Built and tuned Splunk dashboards and detection rules, reducing false positives.
- Ran first-line incident response: evidence collection, timeline reconstruction and reporting.

Signal Officer (Lieutenant), Network Platoon Leader, Republic of Korea Army

Mar 2018 - Dec 2021

- Led a 25-person platoon running military network infrastructure. Ranked #1 of 24 signal sites in a corps-level readiness evaluation.
- Designed and tested network failover and disaster recovery procedures.

SKILLS

Kubernetes: Helm, K3s, EKS, StatefulSets, NetworkPolicy, RBAC, Pod Security

IaC and GitOps: Terraform, Argo CD, GitHub Actions

Observability: Prometheus, Alertmanager, Grafana, Indmon, blackbox exporter, SLOs and burn-rate alerting

Bitcoin and Lightning: LND, Aperture and L402, Neutrino, Bitcoin Core (regtest), channel and fee policy, Static Channel Backups

Cloud and security: AWS (IAM, VPC, EKS, ECR), Vault, software supply chain security

Languages: Go, Python **Data:** PostgreSQL **Spoken:** English (professional working), Korean (native)

CERTIFICATIONS & EDUCATION

Kubernetes: CKA, CKS, CKAD, KCNA, KCSA (Kubestronaut)

AWS: Solutions Architect - Professional, Security - Specialty, CloudOps Engineer - Associate

M.S. Cyber Defense, Dakota State University (2026 - 2028) | **B.S. Civil Engineering, Korea Military Academy** (2014 - 2018)